

ECON 101: Macroeconomics

Solutions – Quiz 3

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Part A: Multiple Choice Solutions

1. **Correct answer: (C)** Greater technological innovation and productivity growth.

Steps/Logic:

- Long-run growth in real GDP per person is driven mainly by **productivity** (output per worker/hour).
- Sustained productivity growth comes from **technological progress**, better organization, and accumulation of knowledge/human capital.
- (A) government consumption does not necessarily raise productive capacity; (B) reducing human capital hurts productivity; (D) lower participation reduces labour input and potential output.

2. **Correct answer: (B)** Real GDP per hour worked.

Steps/Logic:

- Labour productivity measures how much output is produced per unit of labour input.
- In macro, labour input is commonly measured as **hours worked**.
- Therefore: Labour productivity = $\frac{\text{Real GDP}}{\text{Hours worked}}$.

3. **Correct answer: (C)** Increase interest rates and crowd out private investment.

Steps/Logic:

- A budget deficit means government needs to borrow more.
- In the loanable funds market, government borrowing increases **demand** for loanable funds.
- Higher demand (with supply unchanged) raises the **equilibrium real interest rate**.

- A higher interest rate reduces private borrowing for investment: this is **crowding out**.

4. **Correct answer: (B)** Channel savings efficiently into productive investment.

Steps/Logic:

- Financial intermediaries (banks, credit unions, etc.) reduce transaction costs, screen/monitor borrowers, and allocate funds toward productive projects.
- This improves the efficiency of turning savings into investment, supporting capital accumulation and productivity growth.
- (A) intermediaries do not eliminate cycles; (C) and (D) are not primary roles.

5. **Correct answer: (B)** Investment equals national saving (closed economy).

Steps/Logic:

- In a closed economy, there are no international capital flows.
- The resources available for investment come from **national saving**.
- Hence the accounting identity: $S = I$ in a closed economy.

Part B: Numerical Problem Solution

Given:

$$Y = C + I + G$$

$$Y = 5,200, \quad C = 3,300, \quad G = 1,200$$

(All values in billions of dollars.)

(a) Calculate investment I (3 marks)

Step 1: Start from the national income identity.

$$Y = C + I + G$$

Step 2: Rearrange to solve for I .

$$I = Y - C - G$$

Step 3: Substitute the given values.

$$I = 5,200 - 3,300 - 1,200 = 700$$

Answer:

$$I = 700 \text{ (billion dollars)}$$

(b) Compute national saving S (3 marks)

Key idea: In a closed economy, $\boxed{S = I}$.

(Reason: total saving must finance total investment when there is no foreign borrowing/lending.)

Step 1: Use the result from part (a).

$$I = 700$$

Step 2: Apply the closed-economy identity.

$$S = I = 700$$

Answer:

$$\boxed{S = 700 \text{ (billion dollars)}}$$

(Instructor note: If students know the decomposition $S = (Y - T - C) + (T - G)$, they would need T . Since taxes are not provided, the intended method is the closed-economy identity $S = I$.)

(c) Government spending rises by 200; taxes unchanged (4 marks)

Government spending increases:

$$\Delta G = +200$$

Taxes are unchanged, so any change in the government budget balance comes from G only.

Step 1: Effect on national saving.

- Government saving is $(T - G)$.
- If G rises and T is unchanged, then $(T - G)$ falls.
- Therefore national saving S falls.

In symbols (with unchanged T):

$$\Delta(T - G) = -\Delta G = -200$$

So:

$$\boxed{\Delta S = -200}$$

(i.e., national saving decreases by \$200 billion.)

Step 2: Loanable funds market implication.

- The **supply** of loanable funds is national saving S .
- A fall in S shifts the **supply curve of loanable funds left**.

Step 3: Effect on the equilibrium real interest rate.

- With supply lower and demand unchanged, the equilibrium real interest rate rises.

Equilibrium real interest rate increases.

Step 4: Crowding out (intuition, consistent with Ch. 6/7).

- Higher interest rates reduce private investment spending.
- This is the **crowding out** effect.

Final conclusions for (c):

S decreases by 200, r rises, and private I is crowded out.

End of Solution Key